

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0204

Fielder Creek 100 kV
April 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0204. The Interconnection Customer proposes a 179.9 MW natural gas combined cycle facility interconnecting at the Fielder Creek 100 kV substation in the Illinois area. The request was received on February 05, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 153 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Segs 5 to Hillcrest 100kV L: 237.0% loading (129 MVA) following the outage of the Dekalb to Richland 500kV; DFAX = 7.5%.

The Wayne Trace to Darlington Rd 100kV is overloaded to 202.3% of its 111 MVA emergency rating for the loss of the Dekalb to Richland 500kV. The project contributes 6.3% DFAX (11.3 MW) to this overload.

Crry Road Substation to Fielder Crk 100kV: 190.7% loading (115 MVA) following the outage of the Fielder Crk to Deltrail 100kV; DFAX = 57.1%.

For the loss of Richland to Dekalb 500kV, the Slocum Rd to Dekalb 230kV loads to 188.4% of its 508 MVA rating. Project DFAX: 26.9%.

For the loss of Dekalb to Richland 500kV, the Slocum Rd to Mcneil 230kV loads to 173.7% of its 536 MVA rating. Project DFAX: 26.9%.

For the loss of PuebloXtra Substation to Boca Raton 500kV Line, the Pueblo Xtra 500/230 kV Transformer loads to 170.1% of its 965 MVA rating. Project DFAX: 43.5%.

Hill Country 138 230 kV/161 kV Autotransformer: 169.4% loading (100 MVA) following the outage of the Deltrail to Fielder Crk 100kV L; DFAX = 24.8%.

Curry Rd to W Amesbury 100kV L: 168.2% loading (126 MVA) following the outage of the Deltrail to Fielder Crk 100kV; DFAX = 57.1%.

Deltrail to Fielder Crk Substation 100kV: 161.8% loading (165 MVA) following the outage of the Fielder Crk to Curry Rd 100kV Line; DFAX = 64.2%.

The Montibello to Ligonier 100kV Line is overloaded to 151.0% of its 136 MVA emergency rating for the loss of the Dekalb to Jones Chapel 500kV Line. The project contributes 5.6% DFAX (10.1 MW) to this overload.

Cypress Creek to Big Oaks 100kV line: 150.4% loading (136 MVA) following the outage of the Dekalb to Jones Chapel 500kV; DFAX = 5.6%.

Mcneil to Breed 230kV: 150.2% loading (638 MVA) following the outage of the Richland to Dekalb 500kV; DFAX = 26.9%.

For the loss of Dekalb to Richland 500kV, the Breed to Sidney A Murray Jr Hydroelectric 230kV loads to 150.2% of its 638 MVA rating. Project DFAX: 26.9%.

The Rex Brown to Montibello 100kV is overloaded to 148.4% of its 132 MVA emergency rating for the loss of the Dekalb to Jones Chapel 500kV. The project contributes 5.6% DFAX (10.1 MW) to this overload.

The Cypress Crk to Ligonier 100kV is overloaded to 148.0% of its 132 MVA emergency rating for the loss of the Dekalb Substation to Jones Chapel Substation 500kV Line. The project contributes 5.6% DFAX (10.1 MW) to this overload.

Kendallville to Rex Brown Substation 100kV: 147.7% loading (123 MVA) following the outage of the Dekalb to Jones Chapel 500kV; DFAX = 5.6%.

Farmersville to Kendallville 100kV line: 145.3% loading (118 MVA) following the outage of the Dekalb to Jones Chapel 500kV; DFAX = 5.6%.

For the loss of Richland to Troy Energy Llc 500kV line, the Sta S Van Nuys to Minnehaha Substation 230kV loads to 141.6% of its 483 MVA rating. Project DFAX: 17.8%.

The Richland 500/230kV Transformer Bank is overloaded to 140.9% of its 853 MVA emergency rating for the loss of the Troy Energy Llc to Richland 500kV. The project contributes 34.1% DFAX (61.4 MW) to this overload.

Ward Gulf to Laureles 100kV Line: 140.2% loading (175 MVA) following the outage of the Dekalb to Joes Chapel 500kV; DFAX = 5.6%.

For the loss of Boca Raton to Pueblo Xtra 500kV line, the Bluebonnet to Concord 230kV loads to 140.1% of its 787 MVA rating. Project DFAX: 29.7%.

For the loss of Trego to Beaver 500kV, the Herald-Twin Cities to Pickford 230kV line loads to 137.4% of its 664 MVA rating. Project DFAX: 10.5%.

For the loss of Ross Avenue to Deltrail 100kV, the Ce Turbo to Deltrail 100kV loads to 134.7% of its 109 MVA rating. Project DFAX: 32.9%.

For the loss of Dekalb to Richland 500kV line, the Minnehaha to Sidney A Murray Jr Hydroelectric 230kV Line loads to 134.2% of its 545 MVA rating. Project DFAX: 17.3%.

The Robison Park to Dekalb 230kV line is overloaded to 133.9% of its 985 MVA emergency rating for the loss of the Dekalb Substation to Jones Chapel 500kV. The project contributes 35.8% DFAX (64.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Segs 5 to Hillcrest 100kV line	Rebuild line with higher-capacity conductor	\$2,183,000	5% (\$108,000)
Wayne Trace to Darlington Rd 100kV	Rebuild line with higher-capacity conductor	\$2,575,000	5% (\$127,000)
Fielder Crk to Curry Rd 100kV	Rebuild line with higher-capacity conductor	\$7,961,000	48% (\$3,812,000)
Slocum Rd to Dekalb 230kV	Rebuild line with higher-capacity conductor	\$31,669,000	5% (\$1,564,000)
Slocum Rd to Mcneil 230kV	Rebuild line with higher-capacity conductor	\$8,575,000	5% (\$424,000)
Pueblo Xtra 500/230kV XFMR	Replace transformer with higher-rated unit	\$26,862,000	5% (\$1,324,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hill Country 138 230/161 kV TX	Replace transformer with higher-rated unit	\$10,590,000	35% (\$3,695,000)
Curry Rd to W Amesbury 100kV L	Rebuild line with higher-capacity conductor	\$9,821,000	48% (\$4,703,000)
Deltrail to Fielder Crk Substation 100kV	Rebuild line with higher-capacity conductor	\$4,128,000	48% (\$1,976,000)
Ligonier to Montibelo 100kV	Rebuild line with higher-capacity conductor	\$6,432,000	5% (\$318,000)
Cypress Creek to Big Oaks 100kV line	Rebuild line with higher-capacity conductor	\$6,608,000	5% (\$327,000)
Mcneil to Breed 230kV	Rebuild line with higher-capacity conductor	\$25,006,000	5% (\$1,235,000)
Sidney A Murray Jr Hydroelectric to Breed 230kV line	Rebuild line with higher-capacity conductor	\$27,363,000	5% (\$1,351,000)
Rex Brown to Montibello 100kV	Rebuild line with higher-capacity conductor	\$4,494,000	5% (\$222,000)
Cypress Crk to Ligonier 100kV	Rebuild line with higher-capacity conductor	\$6,893,000	5% (\$341,000)
Kendallville to Rex Brown 100kV	Rebuild line with higher-capacity conductor	\$2,727,000	5% (\$135,000)
Farmersville to Kendallville 100kV line	Rebuild line with higher-capacity conductor	\$5,173,000	5% (\$256,000)
Sta S Van Nuys to Minnehaha Substation 230kV	Rebuild line with higher-capacity conductor	\$10,079,000	5% (\$499,000)
Richland 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$32,583,000	5% (\$1,614,000)
Ward Gulf to Laureles 100kV Line	Rebuild line with higher-capacity conductor	\$6,809,000	5% (\$337,000)
Concord to Bluebonnet 230kV	Rebuild line with higher-capacity conductor	\$14,859,000	5% (\$735,000)
Pickford to Herald-Twin Cities 230kV line	Rebuild line with higher-capacity conductor	\$54,008,000	5% (\$2,706,000)
Deltril to Ce Turbo 100kV	Rebuild line with higher-capacity conductor	\$28,137,000	48% (\$13,453,000)
Minnehaha to Sidney A Murray Jr Hydroelectric 230kV Line	Rebuild line with higher-capacity conductor	\$29,871,000	5% (\$1,474,000)
Robison Park to Dekalb 230kV line	Rebuild line with higher-capacity conductor	\$37,608,000	5% (\$1,858,000)
	Total allocated to project		\$44,594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.